

Validator-Guided Repair on a Shared Initial LLM Candidate: A Preregistered Paired Evaluation for Network Configuration Safety

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired confirmatory experiment (PREREG-CAND-001-VER-GVR-2026-09-01) to measure whether permitting a deterministic network-configuration validator plus one automated repair changes the rate of validator-valid executable configurations relative to leaving the identical sampled LLM candidate unguarded (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` (hosted adapter `openai_responses`) and deterministic `netops_validator_v5` on $N = 200$ paired intents (`completed_episode_count = 400`; `model_calls_used = 200`), every shared initial candidate passed validator checks and no repairs were attempted or applied (`repair_attempt_count = 0`; `repair_applied_count = 0`). Observed outcomes: `baseline_success = 200/200`, `guarded_success = 200/200`; paired contingency $n11 = 200$, $n10 = 0$, $n01 = 0$, $n00 = 0$; exact McNemar two-sided $p = 1.0$; paired bootstrap percentile 95% CI (seed = 1729; 10,000 resamples) = [0.0, 0.0]. We present artifact-grounded methodology, execution accounting, representative validator diagnostics, compact contingency and repair summaries, and a focused discussion clarifying the causal limits of the executed estimand under `shared_initial_candidate` semantics and preregistered follow-on designs required to measure repair efficacy under independent sampling, higher difficulty, or adversarial inputs.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration sequences in network operations (NetOps). Erroneous sequences, syntactic mistakes, or fabricated fields can cause service disruption or policy violations when applied to devices. A common architectural mitigation is a generate→validate→repair (GVR) pipeline: generate candidate configuration(s), deterministically validate them against intent/topology/workflow constraints, and trigger automated repair when the validator finds faults. Prior work documents verifier-guided improvements in infrastructure-as-code and related code-generation domains and recommends grounding strategies to reduce hallucination in operational tasks [1–3]. However, whether verifier-guided pipelines reduce execution-level operational risk for network configurations remains an open empirical question because prior demonstrations often measure textual correctness or IaC test suites rather than executed device-state safety in packet-level or emulator contexts [3,4].

This paper reports a fully preregistered paired confirmatory experiment that isolates the downstream causal contribution of a deterministic validator and any permitted repair acting on the

exact same sampled LLM candidate (`shared_initial_candidate` semantics). Under these semantics baseline and guarded episodes for each intent begin from the identical initial generation; any paired difference can therefore only arise from deterministic validation and a permitted repair of that same candidate. Our primary research question is: under `shared_initial_candidate` semantics, does permitting a deterministic validator plus at most one automated repair change the proportion of validator-valid executable configurations relative to leaving the same initial candidate unguarded? The contribution is an artifact-grounded report of the executed estimand with complete execution accounting, exact inferential statistics, representative validator diagnostics, and precise recommendations for preregistered follow-on designs that can measure repair efficacy when independent sampling, elevated difficulty, or adversarial inputs are employed.

II. RELATED WORK

The literature motivating verifier-guided pipelines for operational automation spans documented LLM unreliability, grounding/RAG methods, verifier-guided code/IaC repair systems, and NetOps capability studies. Surveys and empirical analyses consistently identify hallucination and factual unreliability as core risks when applying LLMs to operational tasks; these works motivate grounding and verification layers to limit unsupported or fabricated outputs in high-stakes contexts [1]. Evidence-grounded and retrieval-augmented generation (RAG) designs have been shown to reduce hallucination for operations-style question answering and document retrieval tasks in applied evaluations, improving textual factuality and traceability in AIOps and DevOps scenarios; however, these studies typically measure document/answer correctness rather than network-state executability after applying generated device commands [2].

Generate→validate→repair architectures are an active research direction in adjacent domains. Recent work on infrastructure-as-code (IaC) demonstrates that integrating verifier feedback into generation or fine-tuning loops can reduce failing IaC deployments and improve policy compliance in that domain; these method papers provide a conceptual and empirical foundation for applying verifier-guided repair to other configuration tasks but do not by themselves establish transfer to device-level NetOps execution [3]. Related program-repair and verifier-guided self-correction studies

further show that a verification step plus automated repair can raise textual/program correctness metrics, again typically measured against test suites or formal specifications rather than emulator/execution outcomes. We therefore treat these results as methodological precedent but not as direct evidence that GVR pipelines will prevent unsafe executable network states when generator outputs are enacted on device models or emulators.

NetOps-specific evaluations have studied intent translation and LLM capability for configuration generation, showing promise for mapping operator intent to device commands but emphasizing accuracy metrics and held-out textual correctness rather than standardized execution-level safety or emulator-based validation [4]. Broad AIOps surveys and reviews highlight a persistent gap: many capability benchmarks and capability-oriented evaluations do not include standardized execution-aware safety metrics, and they call for improved validator fidelity and execution-level benchmarks to bridge generation to operational safety [5].

Putting these lines of work together suggests two cautions that guided our design. First, verifier-guided improvements observed in IaC and program-repair domains motivate testing the same architecture in NetOps but do not substitute for execution-level experiments: transferability is an empirical question. Second, experiments that conflate generator sampling variability with downstream validator/repair effects make causal attribution difficult; a clearly specified estimand and sampling semantics are necessary to isolate whether validation/repair changed outcomes for the same generated candidate versus whether improved first-pass sampling reduced errors. These considerations led us to the preregistered shared_initial_candidate paired design used in this study, which isolates downstream validator/repair effects on identical generator outputs while acknowledging that independent-sampling designs are needed to evaluate generator-level error reduction.

III. METHODOLOGY

Preregistration and primary estimand. The study preregistration (PREREG-CAND-001-VER-GVR-2026-09-01) froze the primary estimand and analysis before any model calls. Primary estimand: paired_success_rate_difference_guarded_minus_baseline (per-intent paired difference in proportion of validator-valid executable configurations) under shared_initial_candidate semantics. Design: deterministic paired sample of task_count = 200 intents producing planned_episode_count = 400 episodes. Generation semantics: exactly one sampled initial generation per pair (shared) and deterministic reuse by baseline and guarded episodes. Guarded episodes allowed up to one automated repair (maximum_repair_calls_per_task = 1). Primary test: exact two-sided McNemar test; paired bootstrap (seed = 1729; 10,000 resamples) for a percentile 95% CI.

Task bank and contamination controls. A deterministic task generator produced the manifest of 200 synthetic

intent_configuration_repair_v1 tasks. Each manifest entry records initial_state, required_sequence, difficulty annotations, and a generation_seed for deterministic replay. Preexecution contamination checks (SHA-256 exact-match and n-gram similarity thresholds) were applied; no exact-match contamination exclusions were flagged and the confirmatory sample executed intact. Contamination and task manifest artifacts are archived in the execution evidence bundle.

Model calls, generation semantics, and provider audit. Execution used the declared generator labeled gpt-5-mini via the hosted adapter openai_responses and consumed model_calls_used = 200 (one shared initial generation per pair). The archived model_configuration.json records declared_model_version = "gpt-5-mini" and temperature = null/unset; we explicitly state that null/unset is not evidence of deterministic sampling and does not imply temperature = 0. Provider audit traces record hosted_model_call_event_count = 200, cache_miss_count = 200, and stage_counts showing shared_initial_generation = 200 consistent with the preregistered shared_initial_candidate semantics.

Deterministic validator and repair policy. Scoring used deterministic_netops_validator_v5 with scoring_rule = "full_intent_constraint_validation." The validator parses candidate commands, applies task workflow_policies and required_sequence checks, and produces structured validator traces with validation_before and validation_after subtables, parsed_command_count, required_command_count, normalized_configuration, violation_codes, and a repair_applied flag. For guarded episodes one automated repair attempt was permitted; the repair, if invoked, would produce a new candidate which the validator would re-validate. Under shared_initial_candidate semantics only the guarded arm could alter the identical initial candidate via the permitted repair; the baseline arm reused the same candidate unchanged.

Mapping to outcomes and missingness. Validator.valid = true mapped to a binary "executable/valid" outcome for each episode. Primary analysis adhered to the preregistered complete_pair_primary policy (exclude incomplete pairs). A sensitivity analysis (failure-as-zero) was preregistered to treat missing or failed episodes as unsafe. Multiplicity: single confirmatory McNemar test; exploratory subgroup analyses (e.g., by difficulty patterns) were labeled exploratory and corrective procedures (Benjamini–Hochberg) were specified if applied.

IV. RESULTS

Execution accounting and deterministic reconciliation. The run completed exactly as preregistered: planned_episode_count = 400, completed_episode_count = 400, failed_episode_count = 0, and model_calls_used = 200 (one shared initial generation per pair). Deterministic reconciliation artifacts confirm provider and stage accounting: hosted_model_call_event_count = 200, completed_hosted_model_call_event_count = 200, cache_miss_count = 200, and stage_counts = {shared_initial_generation: 200}; cross_condition_cache_key_reuse_observed = false. No

infrastructure retries or failures were recorded and the analysis missingness_summary reports complete_pairs = 200 with zero failed or unscorable episodes. The analysis executor used paired bootstrap settings bootstrap_resamples = 10000 and bootstrap_seed = 1729 as archived in analysis/analysis_log.jsonl.

Primary outcomes and contingency. Aggregated condition summaries (analysis/condition_summary.csv) show baseline_successes = 200/200 and guarded_successes = 200/200 (success_rate = 1.0 for both conditions). The preregistered paired 2×2 contingency (analysis/paired_contingency_table.csv) is n11 = 200, n10 = 0, n01 = 0, n00 = 0. Exact McNemar two-sided test yields $p = 1.0$, and the paired bootstrap percentile 95% CI for the paired difference (seed = 1729; 10,000 resamples) is [0.0, 0.0]. These results are deterministic given the archived inputs and the shared_initial_candidate semantics: because every shared initial candidate validated successfully, the guarded repair path was never exercised and no paired discordance could occur.

Repair activity and per-episode validator diagnostics. Across the confirmatory sample repair_attempt_count = 0 and repair_applied_count = 0: the validator raised no violations on any shared initial candidate and therefore the permitted guarded repair flow never produced a revised candidate. Per-episode scoring JSON artifacts include full validator traces; representative examples archived under execution/scoring/ (e.g., task-000004-baseline.json, task-000004-guarded.json) illustrate the typical trace structure: shared_initial_candidate text, validation_before and validation_after subtables, parsed_command_count, required_command_count, normalized_configuration, validator_id = deterministic_netops_validator_v5, validator_version = 5.0, and validator.valid = true. For example, task-000004 (representative) had parsed_command_count = 4 and required_command_count = 4; its normalized_configuration was:

```
interface edge2 admin down
interface edge2 vlan 40 interface edge2 admin up
interface edge1 admin down
```

Similarly, task-000005 showed parsed_command_count = 3 with required_command_count = 3 and validator.valid = true; task-000006 showed parsed_command_count = 6 with required_command_count = 6 and validator.valid = true. In each representative JSON the validation_before and validation_after subtables are identical, and repair_applied is false. These per-episode diagnostics demonstrate that, for the sampled tasks and the single sampled candidate per pair, syntactic parsing, required-sequence checks, and workflow_policy constraints were satisfied in the first pass.

Provider audit, contamination, and archived artifacts. Preexecution contamination checks produced no exact-match exclusions; analysis/contamination_summary.csv is empty. Provider audit artifacts and the model_configuration.json (archived) record declared_model_version = "gpt-5-mini" and temperature = null/unset; as stated in Methods, null/unset does not imply determinism. Key archived

analysis files enabling independent verification include execution/raw_results.jsonl, execution/task_manifest.jsonl, analysis/paired_contingency_table.csv, analysis/condition_summary.csv, analysis/missingness_summary.csv, analysis/contamination_summary.csv, and analysis/analysis_log.jsonl. The analysis log documents the bootstrap parameters used for interval estimation and that the paired analysis completed without error.

Compact repair summary and degenerate pathway explanation. Table 2 (preregistered reviewer request summary) explicitly lists total_pairs = 200, repair_attempt_count = 0, and repair_applied_count = 0 (derived from execution/scoring/artifacts and archived analysis files). The degenerate contingency (all pairs valid in both arms) is fully explained by these archived diagnostics: because no validator violations were detected on any shared initial candidate, the guarded pipeline had no opportunity to produce a corrective effect. This artifact-grounded pathway clarifies that the observed null paired difference is a property of the executed estimand (shared_initial_candidate semantics plus the sampled task bank and single generation per pair), not a general statement about the utility of GVR architectures.

Availability of representative traces for replay and further inspection. Every scored episode includes a structured JSON validator trace (validation_before/validation_after, normalized_configuration, parsed_command_count, required_command_count, violation_codes, repair_applied flag) and the shared initial response text. These artifacts permit exact replay of the sampling and deterministic validation steps to confirm that repairs were never invoked and to inspect per-task difficulty annotations present in the task manifest. The archived reconciliation and provider audit files enable independent verification of the execution accounting claims reported above.

V. DISCUSSION

Interpreting the executed estimand. The shared_initial_candidate semantics were intentionally chosen to isolate the causal effect of deterministic validation and any permitted repair acting on the exact same generator output. The executed estimand therefore answers a narrowly defined causal question: when baseline and guarded episodes begin from the identical sampled candidate, can deterministic validation and at most one automated repair change the validator-valid/executable outcome? The observed null effect (paired difference = 0.0) shows that, for this task bank and this single sampled candidate per pair, the validator found no violations and the permitted repair path was never exercised. Importantly, this does not evaluate whether guarded pipelines reduce first-pass generator error rates under independent sampling or different model temperatures; such claims would require a different preregistered estimand and execution.

Threats to validity (artifact-grounded). Internal validity: the paired shared_initial_candidate design eliminates sampling variability across arms and thus preserves attribution of

any observed repair effect to validator/repair alone; however, this design precludes measurement of generator-level effects. Construct validity: `deterministic_netops_validator_v5` enforces syntactic and workflow-policy checks and maps results to a boolean valid flag; it is an abstraction that does not execute vendor CLI on physical hardware nor perform packet-level emulation—dynamic protocol/timing effects are therefore not represented. External validity: results are conditioned on one declared generator (`gpt-5-mini` via adapter `openai_responses`), a synthetic deterministic task bank, and one validator implementation; generalization beyond these artifacts is unsupported. Conclusion validity: zero discordant outcomes preclude inference about repair benefit magnitude under different sampling regimes because the guarded path was not invoked.

Operational implications and preregistered follow-on designs. The null result here should not be read as evidence that GVR pipelines are unnecessary in practice. Instead it demonstrates an estimand boundary: under `shared_initial_candidate` semantics and the executed task bank the guarded pipeline had no opportunity to change outcomes. To empirically evaluate repair efficacy in operationally relevant regimes, we preregistered and recommend the following estimand/design modifications (each reuses archived artifacts where applicable): (A) remove `shared_initial_candidate` semantics so baseline and guarded arms receive independent first-pass samples (this permits the repair path to act on generator faults but requires a revised paired/unpaired analysis plan to separate generation variability from repair effects); (B) select or synthesize tasks labeled "extreme" difficulty or inject adversarial perturbations to raise expected validator failure rates and thereby exercise repair logic; (C) preregister non-null temperature values and temperature sweeps to evaluate sample diversity versus validator-failure tradeoffs; (D) increase validator fidelity by integrating vendor CLI semantics or packet-level emulation so repairs are evaluated against dynamic runtime behavior; (E) expand to multi-model sampling to assess generality across generator families.

Reproducibility and artifact utility. All per-episode scoring JSON artifacts include validator traces with `validation_before/validation_after` subtables, `normalized_configuration`, `parsed_command_count`, `required_command_count`, and `repair_applied` flags permitting independent verification that repairs did not run. Execution and analysis artifacts (`execution/raw_results.jsonl`, `execution/task_manifest.jsonl`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`) are archived in the evidence bundle and enable replay of the exact `shared_initial_candidate` sampling. The `model_configuration.json` records `declared_model_version = "gpt-5-mini"` and `temperature = null`; `null/unset` must not be interpreted as deterministic sampling. The master prompt immutable reference is archived (see Disclosure). These artifacts are sufficient to reproduce the executed estimand and deterministic reconciliation.

VI. LIMITATIONS

- `Shared_initial_candidate` semantics: the design produced exactly one sampled initial generation per intent and reused it across baseline and guarded conditions; this measures validator/repair effects only and cannot detect differences caused by independent per-condition sampling or prompt engineering.
- Temperature unset (`temperature = null`) in the archived model configuration: `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`; sampling nondeterminism may still have occurred during the single sampled generation.
- Single model and single hosted provider: results reflect `gpt-5-mini` under the archived adapter; generalization to other models or model families is unsupported by these artifacts.
- Synthetic task bank and validator abstraction: tasks were synthetic `'intent_configuration_repair_v1'` items and the deterministic validator is an abstraction; realistic vendor CLI idiosyncrasies, emergent runtime interactions, and hardware effects were not executed on live devices.
- No observed validator failures: all initial candidates were validator-valid, so the experiment could not assess the repair step's effectiveness; the null effect therefore reflects the task bank and sampling semantics rather than evidence that GVR is unnecessary in general.

VII. CONCLUSION

We executed a fully preregistered paired confirmatory experiment that measures the downstream effect of a deterministic validator plus an allowed single automated repair on the exact same sampled LLM candidate per intent (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` and `deterministic_netops_validator_v5` on 200 paired intents, every shared initial candidate passed validation and no repairs were attempted or applied; guarded and baseline outcomes were identical for the executed estimand (paired difference = 0.0; McNemar $p = 1.0$; paired bootstrap CI = [0.0, 0.0]). This artifact-grounded null result demonstrates an estimand boundary: under `shared_initial_candidate` semantics and the executed task bank the guarded pipeline could not be exercised and therefore could not change outcomes. It does not imply that GVR pipelines are unnecessary generally. Preregistered follow-on experiments that (1) remove `shared_initial_candidate` semantics, (2) increase task difficulty or inject adversarial inputs, (3) set explicit sampling parameters, and (4) raise validator fidelity are required to empirically evaluate repair efficacy under realistic sampling variability and adversarial conditions. The archived artifacts and reconciliation files enable exact reproduction of the executed estimand and provide a secure base for precise preregistered extensions.

References [1] A. Alansari and H. Luqman, "Large Language Models Hallucination: A Comprehensive Survey," arXiv preprint arXiv:2510.06265, 2025. [2] B. Zhang, H. Rao, and D. Zhao, "Evidence-Grounded RAG for Cloud-Native DevOps: Hallucination-Resistant AIOps Question Answering

over Private Operations Documents," J. Autom. Cloud-Native Syst., 2024. [3] P. Jana et al., "TerraFormer: Automated Infrastructure-as-Code with LLMs Fine-Tuned via Policy-Guided Verifier Feedback," Proc. ACM Symp. (2026). [4] Y. Miao et al., "An Empirical Study of NetOps Capability of Pre-Trained Large Language Models," arXiv:2309.05557, 2023. [5] L. Zhang et al., "A Survey of AIOps in the Era of Large Language Models," ACM Comput. Surv., 2025.

One-line reiteration of the primary estimand limit: the preregistered primary estimand intentionally used `shared_initial_candidate` semantics; evaluating per-condition independent sampling requires a different preregistration and analysis plan (explicitly recommended above).

DISCLOSURE STATEMENT

Disclosure (concise): The human Research Director selected the broad topic family (Generative AI / LLMs for NetOps) before the autonomy lock. After the autonomy lock the autonomous system performed literature review, experiment selection, preregistration (PREREG-CAND-001-VER-GVR-2026-09-01), code generation, execution, deterministic validation, automated analysis, and manuscript drafting without manual human editing of technical content. Declared model used in execution: `gpt-5-mini` via hosted adapter `'openai_responses'` (execution used `model_calls_used = 200`; archived `model_configuration.json` records `temperature = null/unset`; `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`). Generation semantics executed: `shared_initial_candidate` — exactly one sampled initial generation per intent was produced and deterministically reused for both baseline and guarded conditions. Validator: `deterministic_netops_validator_v5` scored candidates using `'full_intent_constraint_validation'` and a single automated repair iteration was permitted in guarded episodes; in this run the validator flagged no violations and no repairs were applied (`repair_attempt_count = 0`; `repair_applied_count = 0`). Execution accounting: `completed_episode_count = 400` (200 paired intents) completed without preregistration deviations. Master prompt immutable reference: `provenance/master_prompt.txt` — SHA-256: `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acd1582bb39b15ba`. Pre-lock infrastructure development and rehearsal occurred but pre-lock artifacts were not used as empirical evidence in this final run. After the autonomy lock no human manually rewrote technical results, manuscript text, figures, or references.

REFERENCES

- [1] Aisha Alansari, Hamzah Luqman, "Large Language Models Hallucination: A Comprehensive Survey", 2025, 10.48550/arxiv.2510.06265
- [2] Boning Zhang, Hengning Rao, Derek Zhao, "Evidence-Grounded RAG for Cloud-Native DevOps: Hallucination-Resistant AIOps Question Answering over Private Operations Documents", 2024, 10.69987/jacs.2024.40308
- [3] Prithwish Jana, Sam Davidson, Bhavana Bhasker, Andrey Kan, Anoop Deoras, Laurent Callot, "TerraFormer: Automated Infrastructure-as-Code with LLMs Fine-Tuned via Policy-Guided Verifier Feedback", 2026, 10.1145/3786583.3786898
- [4] Yukai Miao, Yu Bai, Li Chen, Dan Li, Haifeng Sun, Xizheng Wang, Ziqiu Luo, Ren, Yanyu, Dapeng Sun, Xiuting Xu, Qi Zhang, Chao Xiang, Xinchu Li, "An Empirical Study of NetOps Capability of Pre-Trained Large Language Models", 2023, 10.48550/arxiv.2309.05557
- [5] Lingzhe Zhang, Tong Jia, Mengxi Jia, Yifan Wu, Aiwei Liu, Yong Yang, Zhonghai Wu, Xuming Hu, Philip S. Yu, Ying Li, "A Survey of AIOps in the Era of Large Language Models", 2025, 10.1145/3746635